

THEME 2 | TEAM CODEXA

Vellore Institute of Technology, Vellore

SMART GUIDED TROUBLESHOOTING ENGINE

TEAM MEMBERS

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TEAM LEAD

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Theme 2 | The Problem

Everyday device complaints are vague, while troubleshooting systems expect precise technical language.

THE USER SAYS

“My phone is slow.”

Users describe symptoms in everyday words.

The same complaint can map to several causes.

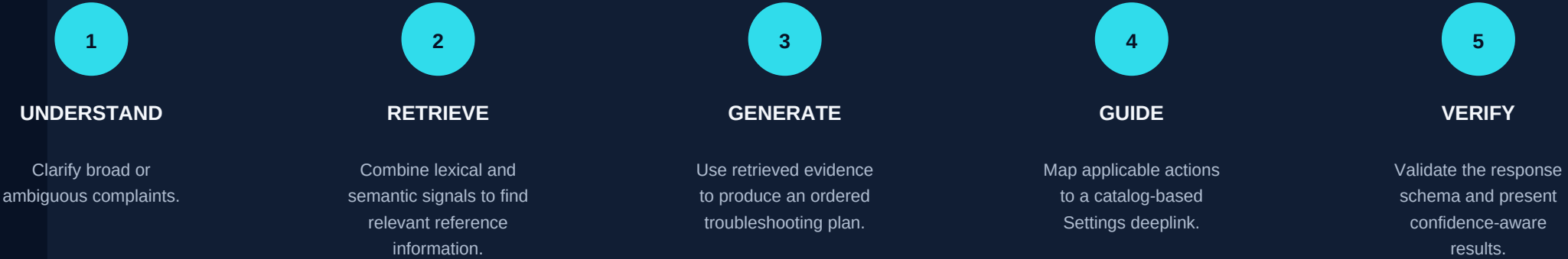
A useful answer must identify the issue, explain what to do, and guide the user to the relevant device setting.

THE GAP

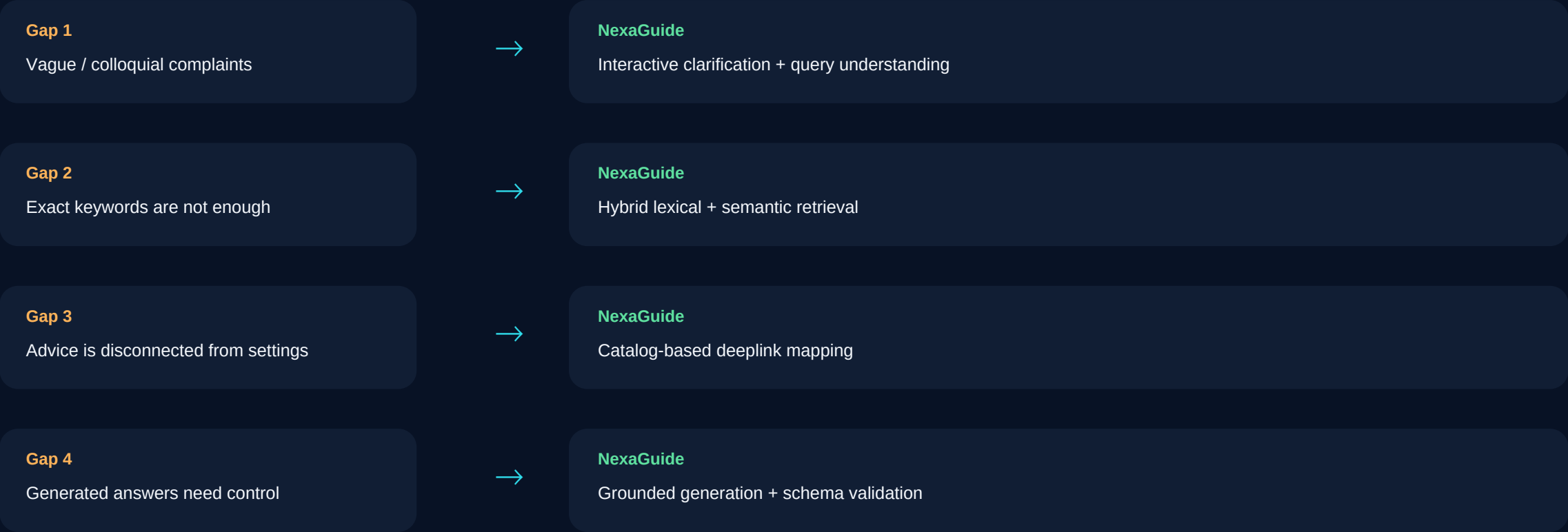
- Keyword-only matching can miss paraphrases.
- Generic answers may not be grounded in reference material.
- Users are told what to do but not always where to go.
- A troubleshooting engine should validate its output.

Our Answer | NexaGuide

A grounded natural-language troubleshooting engine

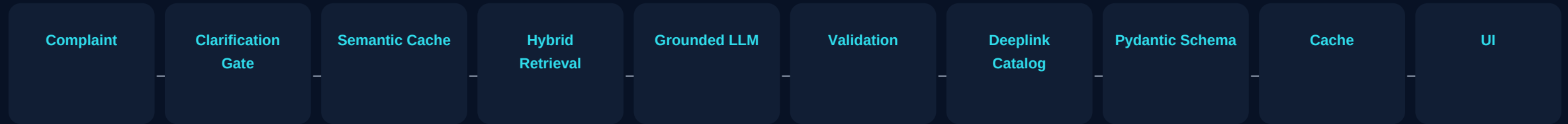


Existing Approaches & How NexaGuide Closes the Gaps



NexaGuide | Solution & Architecture

Actual prototype pipeline from the submitted project



Core engineering components

Interactive clarification for broad complaints

- BM25-style lexical ranking + semantic similarity
- Retrieved reference text supplied to the LLM
- Application validation and structured schema checks
- Semantic-cache fast path for warmed/similar examples

Service interface

GET /health

- POST /v1/clarify
- POST /v1/troubleshoot
- GET /docs
- Web UI for interactive diagnosis

End-to-End Inference Path

Illustrative complaint + structured troubleshooting workflow

USER QUERY

"My screen keeps flickering."

CLARIFY

Check whether the complaint is broad enough to need clarification.

RETRIEVE

Rank the bundled troubleshooting reference using lexical + semantic signals.

GENERATE

Produce structured diagnosis + ordered actions from retrieved evidence.

MAP

Match applicable action to a Settings deeplink from the catalog.

VERIFY

Validate schema and show confidence-aware result.

Prototype Capabilities

What is actually implemented in the submitted NexaGuide package

Interactive clarification

A deterministic clarification layer can ask for more context before troubleshooting.

Grounded planning

The LLM receives retrieved reference text rather than being asked for an unrestricted answer.

Technical inspector

The UI can expose pipeline information such as retrieval and response metadata.

Deeplink catalog

The project ships a catalog of 578 masked Settings deeplink entries.

Hybrid retrieval

Lexical and semantic rankings are fused to narrow the relevant reference information.

Confidence-aware UI

The application exposes an application-level confidence signal.

Semantic cache

Previously warmed or similar examples can use the fast path.

Structured validation

Pydantic-based response validation protects the expected output contract.

Prototype Knowledge Base & Data Scope

Bundled troubleshooting data

20

reference records

The submitted package contains 20 bundled SIIS response records.

The records are used as grounded reference material.

Weak matches can be rejected rather than passed to generation.

Settings mapping

578

catalog entries

The deeplink catalog is shipped with masked placeholder URIs.

Matching is based on description/message/context fields.

Validation deeplinks are paired where available.

Current scope

- Primary bundled examples are focused on device/display and troubleshooting scenarios.
- The package includes test code for response validators.
- The evaluation script measures schema validity, URL leaks, cache hit rate and latency.
- The shipped project does not contain a completed accuracy benchmark table.

Tools & Technology Stack

Frontend

HTML / CSS / JavaScript
NexaGuide web interface

Backend

Python + FastAPI
REST API endpoints

LLM

Groq-hosted LLM
Structured troubleshooting generation

Retrieval

BM25-style lexical ranking
Sentence-transformer semantic embeddings

Validation

Pydantic response schema
Application-level safeguards

Data / Assets

SIIS reference JSON
578-entry deeplink catalog

Runtime

Uvicorn
Local Windows setup + Dockerfile

Testing

Validator tests + evaluation script
Metrics generation support

Innovation, Evidence & Limitations

Claims below are restricted to what is present in the submitted prototype package.

INNOVATION

- Clarification before retrieval
- Hybrid lexical + semantic ranking
- Grounded generation from retrieved reference text
- Action-to-Settings deeplink mapping
- Fast-path semantic cache
- Structured schema validation

EVIDENCE IN THE PACKAGE

- 20 bundled SIIS response records
- 578 deeplink catalog entries
- Validator test file
- Evaluation script
- README with local run instructions
- API routes: health, clarify, troubleshoot, docs

LIMITATIONS

- Knowledge base is development-scale.
- Results depend on the relevance of reference text.
- Live device telemetry is not part of the submitted package.
- Deeplink URIs are masked placeholders.
- The package does not include a completed accuracy benchmark.

Engineering Roadmap

01 EXPAND KNOWLEDGE

Grow scenario coverage and keep symptom → diagnosis → actions → destination mappings auditable.

02 RETRIEVAL QUALITY

Evaluate sparse, semantic and fused retrieval with a labeled benchmark.

03 MODEL CONTRACT

Continue normalizing colloquial text and generating predictable structured JSON.

04 CONTEXT

Use supported device / OS metadata to reduce incompatible Settings mappings.

05 FAST PATH

Pre-validate and cache common scenarios for lower latency.

06 QUALITY LOOP

Track step correctness, latency and failed resolutions for iterative improvement.

Team Codexa | Submission Details

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SUBMISSION

Theme Theme 2 – Smart Guided Troubleshooting Engine

Team Codexa

Institution Vellore Institute of Technology, Vellore

Project NexaGuide – Smart Guided Troubleshooting Engine

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AI disclosure Submitted separately / included in project package

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Signature: _____

NexaGuide

Smart Guided Troubleshooting Engine

Team Codexa | Vellore Institute of Technology, Vellore

THANK YOU